F215 Digital Design — Tutorial 4

4-variable K-maps with don't cares: reduction practice, SOP vs POS, and a real design application

Question 1 — Relationship between MSP and MPS

(a) Identify all prime implicants and essential prime implicants for both the 1s-grouping and the 0s-grouping in the k-map below, then write the minimal SOP (MSP) and minimal POS (MPS).
Is MSP = MPS?

AB\CD	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	0	0	1	1
10	0	0	1	0

(b) Identify all prime implicants and essential prime implicants for both groupings (1s and 0s), then write MSP and MPS.
Is MSP = MPS this time?

AB\CD	00	01	11	10
00	X	1	1	X
01	0	X	X	0
11	0	0	0	0
10	1	0	0	1

(c) Based on (a) and (b): what's actually true in general about the relationship between a minimal SOP and a minimal POS derived from the same don't-care K-map? Can you assume they always agree?

Question 2 — Gray-Coded Input with a Parity Check

A system receives a value X (0–7) encoded as a 3-bit Gray code ($G_2G_1G_0$), along with a 4th parity bit Y. The circuit produces two outputs that are sent to a downstream circuit for further processing, as follows:

- valid, V: 1 if the quartet $G_2G_1G_0Y$ has even parity, 0 if it has odd parity. When V = 0, the downstream circuit does not process its input, it is essentially turned off.
- prime, P: 1 if the represented value X is prime (X = 2, 3, 5, 7); 0 otherwise.

You want to build a minimal circuit for generating V and P based on the requirements above.

(a) Complete the following truth table:

X	$G_2G_1G_0$	Y	V	P
0	000	0		
0	000	1		
1	001	0		
1	001	1		
2	011	0		
2	011	1		

3	010	0		
3	010	1		
4	110	0		
4	110	1		
5	111	0		
5	111	1		
6	101	0		
6	101	1		
7	100	0		
7	100	1		

(b) Using this table: identify the PIs and EPIs for V as a function of (G_2,G_1,G_0,Y), and then write the minimal SOP.

(c) Find the minimal SOP for P as a function of (G_2,G_1,G_0,Y), ensuring a minimal circuit following all requirements. Identify PIs, EPIs and the final expression.

(d) Suppose you forced P = 0 for invalid inputs instead. Derive the SOP again and compare literal counts to (c).

Question 3 — K-map Reduction Practice

For each K-map below (A, B, C, D; X = don't care): first identify all prime implicants and mark which ones are essential, then use them to write the minimal sum-of-products expression.

(a)

AB\CD	00	01	11	10
00	0	0	0	X
01	0	1	1	0
11	0	0	X	0
10	X	1	1	1

(c)

AB\CD	00	01	11	10
00	1	1	1	1
01	0	X	0	X
11	X	1	0	0
10	1	0	0	1

(b)

AB\CD	00	01	11	10
00	0	X	0	X
01	X	0	1	0
11	0	0	0	0
10	1	1	1	1

(d)

AB\CD	00	01	11	10
00	X	0	0	X
01	0	0	1	0
11	1	0	0	0
10	X	1	1	1